

THE SECOND GENERATION OF WEB BASED RAIN GAUGE CALIBRATOR

GENERASI KEDUA DARI KALIBRATOR PENGUKUR CURAH HUJAN BERBASIS WEB

Sensus Wijonarko¹, Tatik Maftukhah¹, Dadang Rustandi², Bernadus Sirenden¹, Mahmudi³, Jalu Ahmad Prakosa¹, Purwowibowo¹

¹ Research Center for Physics

² Development Unit for Instrumentation

³ Research Center for Testing Technology

Email: 19w18r26@gmail.com

ABSTRACT

There are attached bubbles on the cylindrical tube for the first generation of web based rain gauge calibrator (G1 WBRGC, WBRGC version 1, WBRGC V1). These bubbles raise questions from users about the appropriateness of this calibrator. An improvement especially by adding a tailor made funnel does not help to clear up the bubbles. This study was intended to eliminate the bubbles by replacing WBRGC V1 into the second generation of web based rain gauge calibrator (G2 WBRGC, WBRGC version 2, WBRGC V2) using problem solution method where some activities were carried out particularly to design, construct, test, revise, observe bubble presence, search bubbles source, and calibrate. The result shows that there were no bubbles anymore in the cylindrical tube for the WBRGC V2.

Keywords: Bubbles, perception, web based rain gauge calibrator (WBRGC) version 1, problem solution method, WBRGC version 2, and bias

ABSTRAK

Ada gelembung-gelembung yang menempel pada tabung silinder dari kalibrator pengukur curah hujan berbasis web generasi pertama (G1 WBRGC, WBRGC versi 1, WBRGC V1). Gelembung-gelembung ini menimbulkan pertanyaan dari pengguna tentang kelayakan kalibrator ini. Perbaikan terutama dengan menambahkan corong yang dibuat khusus tidak membantu menghilangkan gelembung-gelembung tersebut. Penelitian ini dimaksudkan untuk menghilangkan gelembung-gelembung tersebut dengan cara mengganti WBRGC V1 menjadi kalibrator pengukur hujan berbasis web generasi kedua (G2 WBRGC, WBRGC versi 2, WBRGC V2) dengan menggunakan metode penyelesaian masalah di mana beberapa kegiatan dilakukan, khususnya merancang, membuat, menguji, merevisi, mengamati kehadiran

gelembung, mengamati sumber gelembung, dan mengkalibrasi. Hasil penelitian menunjukkan bahwa tidak terdapat gelembung-gelembung lagi pada tabung silinder untuk WBRGC V2.

Kata kunci: Gelembung, persepsi, kalibrator pengukur curah hujan berbasis web (WBRGC) versi 1, metode penyelesaian masalah, WBRGC versi 2, kesalahan sistematis.

I. INTRODUCTION

Rain water is the life trigger. In the right size, it makes vegetation to emerge and grow up quickly to support further life. Droughts or floods, on the other hand, are generally unbeneficial from economic aspect and dangerous from environment side. Hence, rain gauges and their calibrators are the key for the water management.

From many points of view, WBRGC V1 (Wijonarko et al., 2017) is very competitive to be compared with other rain gauge calibration systems or methods (Wijonarko et al. 2019), namely Static Calibration and Dynamic Calibration methods (Calder & Kidd, 1978), and Automated Dynamic Calibration method (Humphrey et al., 1997). The WBRGC V1 (Figure 1), however, still has a disturbing appearance due to the present of bubbles attached on its cylindrical tube, when it is filled directly with water (Wijonarko et al., 2019; Maftukhah & Wijonarko, 2019). The bubbles stimulate hesitation for the WBRGC V1 accuracy. Hence, this problem should be solved promptly.

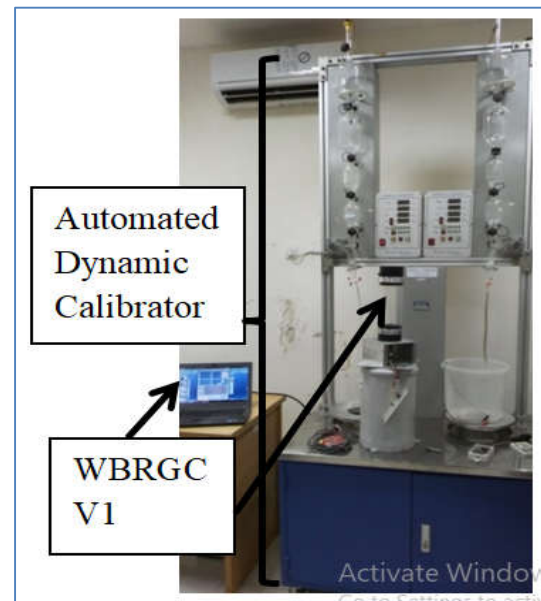


Figure 1. An automated dynamic calibrator and a WBRGC V1

The aim of this study was to overcome the problem by developing a WBRGC V2. The solution effectiveness would be accessed by searching the availability of bubbles using naked eyes. Naked eyes are sufficient way to investigate bubbles because the most important thing in this problem is the user perceptions through their naked eyes. Perception is man's primary form of cognitive contact with the world around him (Efron, 1969). Perception using naked eyes greatly influences the decision.

II. METHODOLOGY

WBRGC V1 was uplink calibrated at an official calibration service office to make unbroken chain to the national standard for measurement unit in this measurand

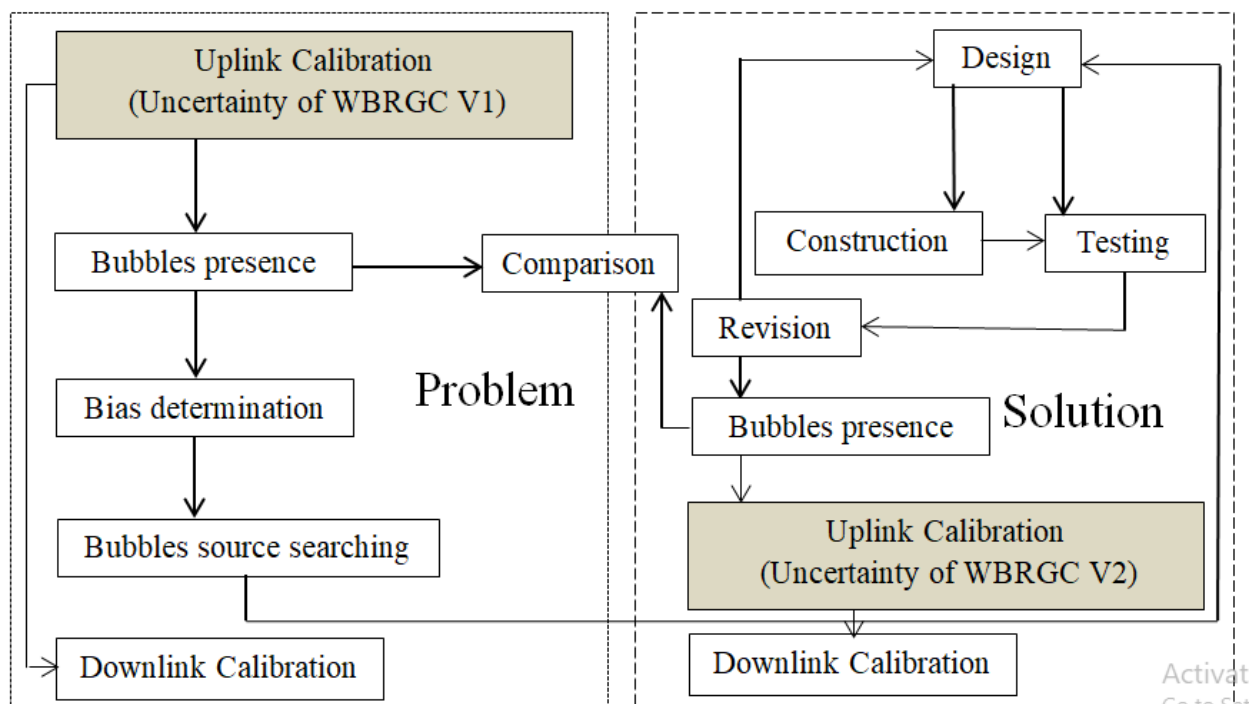


Figure 2. Research method

From this calibration, the uncertainty of WBRGC V1 was 3 ml of 966 ml or around 0.31 % (Wijonarko et al., 2019). Uncertainty is a natural parameter that can be used to characterize any measurement process (Khanam, 2009) or a parameter, associated with the result of a measurement that characterizes the dispersion of the values that could be reasonably attributed to the measurand, namely the particular quantity subject to measurement.

The WBRGC V1 is always equipped with random bubbles attached on

(Figure 2). The uplink calibration is important, but it is beyond the scope of this study. Hence, this activity was submitted to the authorized party in this field, who is expected can give more objective result.

its cylindrical tube. Careful fillings, even with a tailor made funnel, did not clear up the bubbles. Therefore, a new version should be proposed, because the auxiliary tool could not overcome the problem.

Bias due to bubbles was conducted by investigating its value. The bias was around - 0.16 % or around 50 % of the WBRGC V1 uncertainty (Wijonarko et al., 2020). This result reinforces the reason that there was a problem in the WBRGC V1 that needs to be fixed.

Solution would be carried out after the bubbles source had been obtained first.

In this step, an experiment to obtain the source should be accompanied with a literature study to answer why bubbles are always present on the cylindrical tube.

This study concerned to the solution steps. First was the design. The water was inputted from the bottom of cylindrical tube in the WBRGC V2. This tube was designed for 320 ml (Maftukhah et al., 2019), a volume for 10 mm tipping bucket when the diameter of its mouth around 20 cm. This calibrator, however, can be employed to calibrate any dimension.

Second was the construction. Water was inputted or outputted using a pump that can works in two directions. The upper and lower parts of cylindrical tube were not in ink bottle model, but only in the lower part. The upper part was open like a tube.

Third, testing was conducted for design and construction phases. This step was to make sure that WBRGC V2 works correctly.

Fourth, when flaws or defects were obtained, a revision was executed. The revision was conducted several times.

Fifth was the bubble presence observation. Compared to WBRGC V1, the result is sufficient if there is no bubble anymore in the cylindrical tube.

Sixth, WBRGC V2 was uplink calibrated such as WBRGC V1. This step actually is only to fulfill a compulsory for

developing a calibrator, but it is not the main intention for this study.

Seventh was downlink calibration. This step is important for introducing WBRGC to user candidates.

III. EXPERIMENTAL SET UP

Experiments were set up in some steps. The first step was the WBRGC V2 development, where its output is a prototype. The second step was the uplink calibration for both types of calibrators, namely WBRGC V1 (Figure 3a) and WBRGC V2 (Figure 3b). The process and result of second step were beyond the author control, because each uplink calibration in this study was carried out by an official calibration service office. The third step was downlink calibration (Figure 3c and 3d). Downlink calibrations were just to make sure that each WBRGC worked well after had been calibrated. These steps are optional for this study, but they are important for promoting WBRGC to public.

Figure 3 is an integration of simplified experiments carried out at different place and time. The apex activity in this study was to observe bubbles for WBRGC V1 (2a) and WBRGC V2. The development is success when there is no bubble anymore attached on the cylindrical tube.

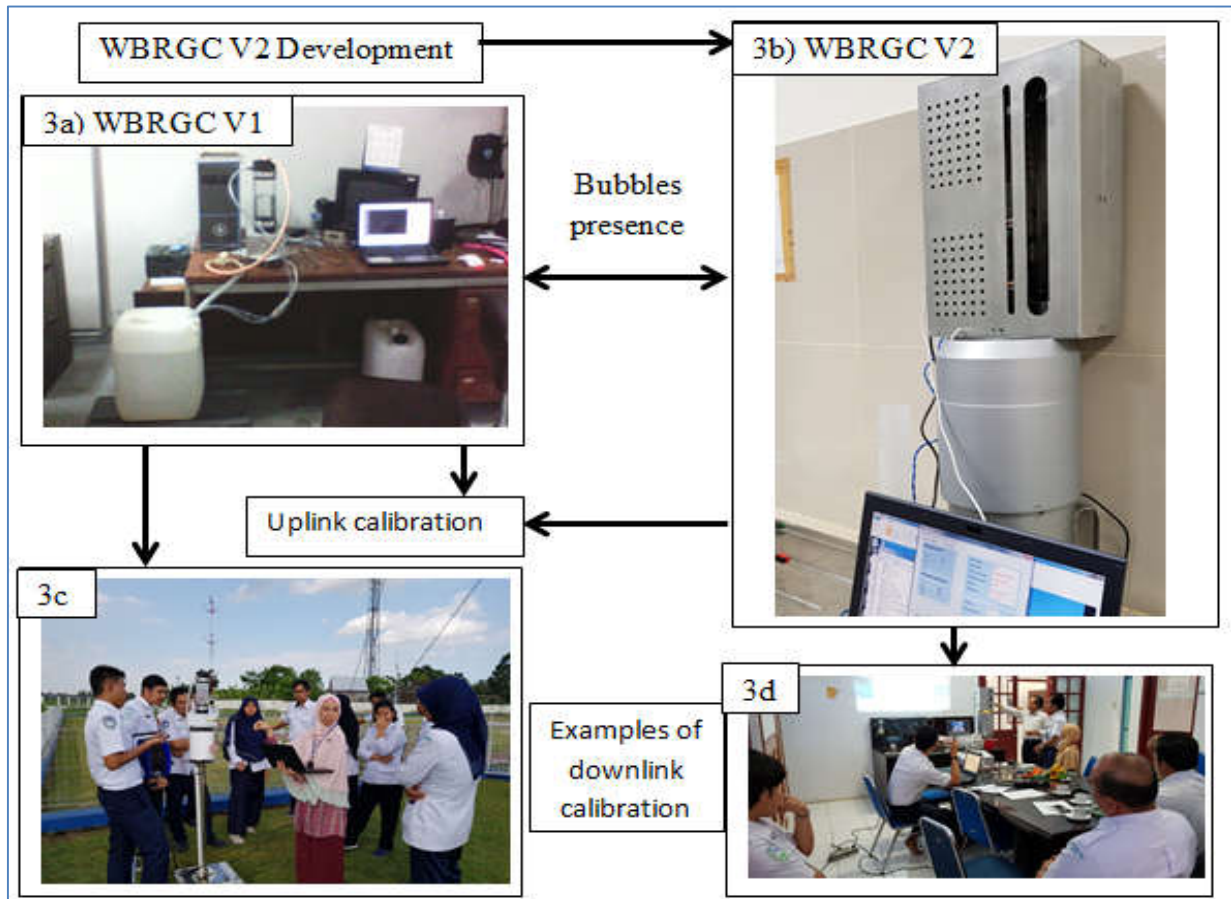


Figure 3. Experimental set up

IV. RESULT AND DISCUSSION

There are some techniques to cope the bubble problem such as the use of funnel (Wijonarko et al., 2020) or electric pulses (van Heesch et al., 1994). The selected technique in this study, however, was the development of second generation WBRGC.

The measurement comparison method (Wijonarko et al., 2016) is rather difficult to be applied as a judgment for the solution effectiveness, because the water volumes of WBRGC V1 and WBRGC V2 are not exactly the same. The simple but effective way is to compare the bubble

presence. There were no visible bubbles in WBRGC V2 anymore. This means that WBRGC V2 has solved the bubble problem that present in WBRGC V1. This work out is supported by some determinant changes in WBRGC V2 (Figure 4).

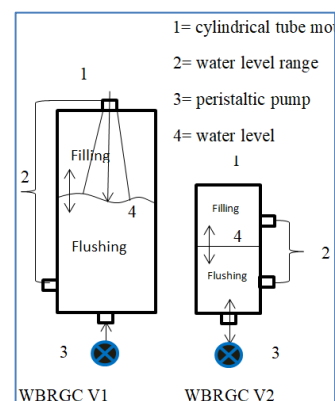


Figure 4. Determinant concept differences between WBRGC V1 and WBRGC V2

First, there is no bottle neck at the inlet of cylindrical tube. The bottle neck or orifice will reduce the flow rate (Davidson & Schöler, 1997), including the flow rate of entrapped air in water. The diameter size of the cylindrical tube in WBRGC V2 is uniform from the lowest water level to the highest water level. This technique will eliminate bubbles in the ceiling of cylindrical tube.

The second change was conducted by limiting the highest and lowest water levels with an optoelectronic sensor respectively. The highest water level is not the top and the lowest water level is not the bottom of the cylindrical tube. The range of calibration water is between some mm lower than the top and some mm higher than the bottom. Hence, there are no bubbles on the highest and lowest water levels.

Third, the peristaltic pump in the WBRGC V1 is used in unidirectional way (downward direction), namely to pour or flush water from the cylindrical tube to the UUT during calibration processes. Peristaltic pump is a mechanical displacement pump that induces flow in a fluid-filled, flexible-walled conduit through peristalsis-transport due to traveling contraction waves (Berg & Dallas, 2015). The peristaltic pump in the WBRGC V2 is employed in bidirectional way, namely in the upward direction -

where it is used to fill water from a small water reservoir to the cylindrical tube - and in the downward direction for calibration processes and water flushing.

Fourth, a bubble formation is caused by impact velocity (Hung et al., 2013). The application of peristaltic pump in the upward direction makes a low impact velocity, because the water for calibration is pushed up in a steady flow. The pushed up water level in a laminar flow makes no entrapped air in the water and no ripples on the water level. A washout phenomenon (Takeuchi & Katase, 1982; Sportisse, 2007) - where in this case is the atmosphere air pushed down by the flowing water from the upper site - does not occur anymore. There is no air surrounded by water. Therefore, no air is rolled up by water to yield bubbles. Briefly, this technique reduces possibilities of air diffusion.

The success of WBRGC V1 development was accompanied with its relative uncertainty result. The relative uncertainty of the WBRGC V1 was 3 ml/966 or 0.31 % (Wijonarko et al., 2019), while the relative uncertainty of the WBRGC V2 (Wijonarko et al., 2019) was 0.5 ml/320 ml or 0.16 %. It means that the relative uncertainty of the WBRGC V2 was about a half of WBRGC V1.

The very low relative uncertainty for WBRGC V2 has a certain advantage. This calibrator can reduce the uplink

calibration frequency, because it has a wide uncertainty interval. As long as it remains inside the interval, no calibration is performed (Carbone, Macii, & Petri, 2003).

In practical point of view, the absence of bubbles is much more significant than the relative uncertainty reduction. The relative uncertainty reduction cannot be seen directly. Even with 0.31 % as WBRGC V1, this relative uncertainty is still very good for calibrating rain gauges. The uncertainty for Class A liquid precipitation gauges is usually $\geq 5\%$ (WMO, 2008; 2012). Hence, the TUR is higher than 16 : 1. It is much better than 10 : 1, the common maximum TUR (Macii & Petri, 2007).

TUR (Test Uncertainty Ratio) is a comparison between the tolerance being checked and the measurement uncertainty; and TAR (Test Accuracy Ratio) is a comparison between the tolerance being checked and the accuracy of measuring equipment and (Mitutoyo, 2018). TAR is also a comparison between the accuracy of the Unit Under Test (UUT) and the accuracy of the standard (Bennett, 2005). From the above statements, it can be inferred that the tolerance in the definition is accuracy.

V. CONCLUSION AND SUGGESTION

Web based rain gauge calibrator V2 was effective solution to handle the presence of

bubbles in the web based rain gauge calibrator version 1. This problem solution method might also be applied for other systems where the bubble presence is unintended. This second generation calibrator, however, still has some aspects to be enhanced.

VI. ACKNOWLEDGEMENT

The funding for this study came from LIPI 217 & 2018 and Kemenristek-Dikti 2019. We also would like to express our sincere thanks to Mr, Surip Kartolo from Research Center for Physics – LIPI for his assistance to our researches.

VII. REFERENCES

- Bennett, K. & Zion, H. (2005). Metrology Concepts: Understanding Test Uncertainty Ratio (TUR). TRANSCAT Calibration & Repair Services.
- Berg J.M., Dallas T. (2015) Peristaltic Pumps. In: Li D. (eds) Encyclopedia of Microfluidics and Nanofluidics. Springer, New York, NY. DOI: https://doi.org/10.1007/978-1-4614-5491-5_1198.
- Calder, I. & Kidd, C. (1978). A Note on the Dynamic Calibration of Tipping-Bucket Gauges. Journal of Hydrology, 39: 383—386.

- Carbone, P. Macii, D. & Petri, D. (2003). Measurement uncertainty and metrological confirmation in quality-oriented organizations. *Measurement*, 34: 263–271
- Davidson, J. F. & Schüler, B. O. G. (1997). Bubble Formation at an Orifice in A Viscous Liquid. *Trans IChemE*, 75: S105-S115.
- Efron, R. (1969). What is perception. In: *Proceedings of the Boston Colloquium for the Philosophy of Science 1966/1968*. <https://doi.org/10.1007/978-94-010-3378-7>.
- Humphrey, M. D., Istok, J. D., Lee, J. Y., Hevesi, J.A., & Flint, A. L. (1997). A New Method for Automated Dynamic Calibration of Tipping-Bucket Rain Gauges. *Journal of Atmospheric and Oceanic Technology*, 14: 1513-1519.
- Hung, Y. L., Wang, M. J., Huang, J. W., Lin, S. Y. (2013). A Study on the Impact Velocity and Drop Size for the Occurrence of Entrapped Air Bubbles-Water on Parafilm. *Experimental Thermal and Fluid Science*, 48: 102-109.
- Khanam, S. (2009). Test Uncertainty Ratio (TUR) and Test Uncertainty. Dissertation.
- Maftukhah, T., Wijonarko, S. (2019). A Data Processing Program for Web Based Rain Gauge Calibrators. *Jurnal Instrumentasi*, 43(1), 53-64. DOI: <http://dx.doi.org/10.31153/instrumentasi.v43i1.178>
- Maftukhah T, Wijonarko S, Sirinden BH, Purwowibowo, Afandi MI, Mulyanto I, Rustandi D. (2019). Alat Kalibrator Sensor Tipping Bucket berbasis Web dengan Variasi Volume Air Tabung dan Metode Penggunaannya. Indonesia Patent No. P00201904061.
- Mitutoyo (2018). Decision Rules, TAR, and TUR. Mitutoyo America Corporation.
- Sportisse, B. (2007). A review of parameterizations for modeling dry deposition and scavenging of radionuclides. *Atmospheric Environment* 41: 2683-2698.
- Van Heesch, E. J. M., Lemmens, R. H. P., Franken, B., Ptasinski, K. J., & Geurts, F. L. S. (1994). Pulsed Corona for Breaking up Air Bubbles in Water. *IEEE Transactions on Dielectrics and Electrical Insulation*, 1(3): 426-431.
- Takeuchi, N. & Katase, A. (1982). Rainout-Washout Model for Variation of Environmental Gamma-Ray Intensity by Precipitation. *Journal of Nuclear Science and Technology*, 19(5): 393-409.
- Wijonarko, S., Rustandi, D., Maftukhah, T., Mahmudi, Sirenden, B.H., Prakosa, J.A., Purwowibowo (2020). The Bias

- of Web Based Rain Gauge Calibrator due to Bubbles. *Jurnal Instrumentasi*, 44(1), 1-12; DOI: <http://dx.doi.org/10.31153/instrumentasi.v44i1.215>.
- Wijonarko, S., Maftukhah, T., Rustandi, D., Sirinden, B. H., Prakosa, J. A., Sediono, W., Darna, Y., Sukanda, R., Pinandito, M. (2017). Sistem Otomatis untuk Mengkalibrasi Pengukur Curah Hujan Tipe Cawan Berjungkit (*Tipping Bucket*). Indonesian Patent No. P00201707655.
- Wijonarko, S., Maftukhah, T., Rustandi, D., Tjahyo, E. D. N., Permana, D., Santosa, B. (2016). A Method to Compare Two Hyetometer Calibrators. *PPI KIM*, 354-364.
- Wijonarko, S., Maftukhah, T., Rustandi, D., Sirenden, B. H., Tjahyo, E. D. N. (2019). Web Based Rain Gauge Calibrator. *Jurnal Instrumentasi*, 43(1), 25-42; DOI: <http://dx.doi.org/10.31153/instrumentasi.v43i1.176>.
- WMO (2008). WMO No. 8: Guide to Meteorological Instruments and Methods of Observation. Geneva.
- WMO (2012). WMO No. 8: Guide to Meteorological Instruments and Methods of Observation, 2008 edition, updated in 2010. Geneva.